

SRPA-YOLO: Shared-Representation and Private-Rank Allocation for Real-Time Small-UAV Detection

Xianpu Wang · Haijie Zhi · Qing Fei

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Abstract Long-range unmanned aerial vehicles (UAVs) occupy few pixels and are readily confused with background clutter, while onboard detection is constrained by memory and frame time. This study introduces Shared-Representation and Private-Rank Allocation YOLO (SRPA-YOLO), a compact one-stage detector that allocates prediction-head capacity according to task requirements. A 64-channel structurally re-parameterized representation is shared by classification and box-distribution prediction, whereas a zero-output-initialized 16-channel classification-private representation contributes an additive classification residual. This allocation makes the box output directly independent of the classification-private representation. Before deployment, the two stems and their task projections are fused algebraically into one 80-channel 3×3 convolution followed by one output projection. The training-time private path is thus fully absorbed into the single-path deployment graph. On the hash-deduplicated DUT Anti-UAV test set, the validation-selected checkpoint achieves 95.850% Precision, 92.201% Recall, 95.185% AP@0.50, 67.071% AP@0.50:0.95, and 77.329% AP@0.75 with 2.164 M deployed parameters. Relative to YOLOv8n, these metrics increase by 3.088, 8.242, 5.057, 6.895, and 9.750 percentage points, respectively, while the parameter count decreases by 0.842 M. Two same-process paired measurements yield latencies of 9.7435 and 9.8574 ms for SRPA-YOLO and 9.8859 and 9.9893 ms for the paired YOLOv8n control. Under the evaluated protocol, the results

indicate that constrained classification-private capacity improves the accuracy–compactness trade-off while retaining comparable frame time.

Keywords unmanned aerial vehicle · small-object detection · real-time image processing · structural re-parameterization · knowledge distillation · lightweight detector

1 Introduction

Small unmanned aerial vehicles (UAVs) are increasingly used in logistics, inspection, emergency response, and low-altitude sensing. The same accessibility also creates a demand for vision-based detection around protected airspace. Compared with generic object detection, long-range UAV detection is unusually unforgiving: a target may cover fewer than 16×16 pixels after letterboxing, motion and defocus weaken its contours, and clouds, birds, buildings, and foliage generate visually similar responses. A detector must therefore raise recall without flooding the scene with false positives, and it must do so under the memory and latency constraints of an edge platform. Recent surveys identify this accuracy–latency tension as a central problem in vision-based anti-UAV systems [32, 25].

One-stage detectors such as YOLO and SSD established direct dense prediction as a practical route to real-time detection [21, 19]. Feature-pyramid and path-aggregation designs subsequently improved the exchange of high-level semantics and fine spatial details [15, 18]. Anchor-free formulations and quality-aware classification further reduced hand-designed priors and aligned confidence with localization quality [24, 16, 14]. Yet simply increasing the number of high-resolution branches or stacking attention modules can increase memory traffic and kernel-launch overhead. EfficientDet, YOLOv4, and YOLOv7 demonstrate that practical speed depends on

X. Wang · Q. Fei
School of Automation, Beijing Institute of Technology, Beijing 100081, China
E-mail: 3220252903@bit.edu.cn
E-mail: feiqing@bit.edu.cn
H. Zhi
College of Energy Engineering, Zhejiang University, Hangzhou 310027, China
E-mail: 22427029@zju.edu.cn

coordinated backbone, neck, training, and deployment design rather than FLOPs alone [23, 2, 27].

The central observation of this study is that classification and localization do not require identical feature subspaces. Classification favors semantic invariance and inter-class discrimination, whereas localization must preserve boundary- and displacement-sensitive information. TSD showed that a common proposal and representation can induce task conflict [29], and TOOD demonstrated the value of combining task interaction with task-specific features and alignment [5]. A fully decoupled head reduces direct feature competition but duplicates costly spatial processing. Conversely, a fully shared head is compact but constrains both tasks to the same representation budget. These considerations motivate a constrained alternative: retain a compact basis for both tasks and allocate a small additional representation only to classification.

We therefore propose SRPA-YOLO, short for Shared-Representation and Private-Rank Allocation YOLO, as a structural modification of the YOLOv8 one-stage framework. YOLOv8n serves as the primary baseline for attributing changes in accuracy, parameter count, and latency. The detector retains P2–P4 feature maps through a lightweight bidirectional neck and applies a structurally re-parameterized head at each scale. A 64-channel shared stem supports box-distribution and class prediction, whereas a 16-channel classification-private stem supplies an additive class residual. Zero initialization of the private output projection preserves the shared detector’s initial input–output function. Before inference, RepVGG-style structural re-parameterization [4] concatenates the equivalent stem kernels and merges their task projections into a single path. Classification capacity is therefore expanded during optimization without retaining a parallel private inference branch.

Training follows two stages. Stage I learns the shared detector from standard supervision and response distillation. Distribution-level localization distillation transfers the teacher’s discrete box distributions [37], while a confidence-weighted term emphasizes reliable teacher locations. Stage II starts from the function-preserving zero-residual state and optimizes only the classification-private stem and projection. Under this constraint, the learned box function remains fixed while classification receives additional representational capacity.

The method combines constrained task-specific representation allocation, function-preserving initialization, and exact deployment fusion; task decoupling, zero initialization, distillation, and structural re-parameterization are established techniques individually. The principal contributions are as follows:

1. We formulate prediction-head capacity as shared and classification-private representation allocation. The resulting 64+16 design makes the box output directly in-

dependent of the private representation, while allowing classification to use the combined shared and private row spaces.

2. We introduce a function-preserving two-stage realization in which a zero-output private projection is refined after the shared detector has converged. This isolates additional classification capacity without changing the established box-regression mapping during Stage II.
3. We derive an exact fusion rule that converts the shared and private training paths into one 3×3 stem and one 1×1 output layer per scale, preserving the computed outputs up to floating-point roundoff and eliminating private-branch inference overhead.
4. A controlled evaluation with validation-selected checkpoints, IoU-resolved AP, matched Stage-II seeds 40/41/42, and same-process paired latency measurement characterizes the resulting detector. Relative to YOLOv8n, the final checkpoint improves Precision, Recall, AP@0.50, mAP@0.50:0.95, and AP@0.75 by 3.088, 8.242, 5.057, 6.895, and 9.750 percentage points, respectively, while reducing deployed parameters by 28.0%.

2 Related Work

2.1 Real-Time and Small-Object Detection

The COCO evaluation protocol established averaged AP over intersection-over-union (IoU) thresholds as a standard measure of localization quality [17]. For small objects, however, repeated downsampling destroys both energy and phase information before a detector can form a stable response. High-resolution pyramids mitigate this loss, but their cost scales with spatial area. YOLOv8 provides a compact anchor-free implementation and reproducible training stack [26]. Because SRPA-YOLO is developed within this framework, YOLOv8n is the primary structural and deployment baseline. YOLO11 is included as a contemporary compact control, whereas UAV-specific reproductions test whether the proposed design remains competitive beyond its parent architecture.

Hardware-aware modules reduce the cost of retaining high-resolution features. MobileNet replaces dense spatial convolution with depthwise separable operations [8]; GhostNet synthesizes redundant features through cheap transforms [6]; FasterNet argues that realized FLOPS and memory access are as important as nominal FLOPs [3]. Our backbone and neck use C2f aggregation, Ghost convolution, depthwise downsampling, and a partial-convolution stage in this spirit. CBAM combines channel and spatial selection [28], while efficient multi-scale attention (EMA) captures cross-spatial relationships with limited parameters [20]. These modules are retained only where the measured architecture benefits; the paper’s novelty is not an attention replacement.

2.2 Anti-UAV Benchmarks and Specialized Detectors

Anti-UAV benchmarks have progressively expanded visible and thermal tracking conditions [35, 11, 9]. The DUT Anti-UAV detection subset is especially suitable for evaluating long-range single-class UAV detection because it contains large background regions and extreme target-scale variation. Specialized detectors have addressed these conditions through stronger feature fusion, lightweight blocks, and attention. YOLOv7-GS modifies the real-time detector for cluttered anti-UAV scenes [1]; EDGS-YOLOv8 and DRBD-YOLOv8 combine lightweight operations with enhanced multiscale representation [10, 12]; RLRD-YOLO emphasizes the balance between a lightweight model and robust small-object recognition [13].

IASL-YOLO combines a lightweight backbone with adaptive feature fusion, whereas BRA-YOLOv10 couples multiscale representation with structured channel pruning [30, 34]. Both methods improve UAV detection by reallocating computation in the backbone or feature pyramid. SRPA-YOLO addresses a complementary source of inefficiency inside the prediction head: it constrains the classification and localization feature row spaces, assigns a low-rank classification-private representation, and removes the training-time decomposition through exact fusion. The distinction is therefore architectural—feature-extraction and pyramid efficiency in the former methods versus task-specific head-capacity allocation and static deployment equivalence in SRPA-YOLO.

Broader aerial benchmarks such as VisDrone contain multiple object classes and denser scenes [38]. Their task semantics differ from the single-class long-range UAV detection problem considered here, so published results on those benchmarks are not directly comparable with the present DUT Anti-UAV evaluation. Evaluation under additional target distributions remains a direction for future work.

2.3 Task Conflict, Distillation, and Deployable Training

Dense heads often share features before separate output layers. TSD and TOOD show that this sharing can create misaligned optima between class confidence and box localization [29, 5]. Our response is not a fully decoupled head; it is a low-rank task allocation in which expensive spatial representation remains mostly shared. The classification-private representation is additive and classification-only, so the box Jacobian with respect to this representation is identically zero.

Knowledge distillation transfers function-space information from a larger teacher to a compact student [7]. Detector-specific methods distinguish foreground/background regions and localization distributions. Attention transfer and focal/global distillation operate on intermediate representations [33, 31], whereas localization distillation treats discrete

regression distributions as structured knowledge [37]. Our final training uses response-level class distillation and dense/focused DFL distillation, but no feature imitation or auxiliary detector at inference.

Structural re-parameterization provides the deployment bridge. RepVGG shows that linear branches with batch normalization can be converted into a single equivalent convolution [4]. We apply the same algebra to a shared/classification-private task decomposition: the stem kernels are concatenated in the channel dimension, the private columns of the box projection are padded with zeros, and the class projections are concatenated. This produces a static deployment graph without gating, dynamic routing, or additional post-processing.

3 Materials and Methods

3.1 Dataset and Evaluation Protocol

The visible-light detection subset of DUT Anti-UAV is used to evaluate long-range single-class UAV detection. The official partition contains 5200 training images, 2600 validation images, and 2200 test images. The original aspect ratio is retained, and every image is letterboxed to 640×640 pixels during training and evaluation. Checkpoints are selected exclusively according to validation mAP@0.50:0.95, after which the selected model is evaluated on the test partition.

Image-level SHA-256 verification identified one duplicated image between the training and test partitions. The source files are retained for traceability, while the duplicated test image is excluded through a fixed evaluation manifest. The resulting test set contains 2199 images and 2244 annotated UAV instances. For scale-resolved analysis, target size is measured after the same letterbox transformation used by the detector. An object is categorized as tiny when its maximum side is below 16 pixels, small when it is between 16 and 32 pixels, and large when it is at least 32 pixels. These intervals reflect the apparent target sizes of the DUT Anti-UAV task.

3.2 Overall Network Architecture

SRPA-YOLO follows a one-stage dense-detection pipeline composed of a multiscale backbone, a lightweight bidirectional neck, and three prediction levels at strides 4, 8, and 16. Figure 1 summarizes the complete data flow and the transformation from the training head to the deployment head.

3.2.1 Multiscale Spatial Feature Extraction

Let $X_0 \in \mathbb{R}^{3 \times 640 \times 640}$ be the input image. At backbone stage l , the feature map is written as

$$X_l = \mathcal{B}_l(X_{l-1}) = \mathcal{A}_l(\text{BN}(K_l * X_{l-1} + b_l)), \quad (1)$$

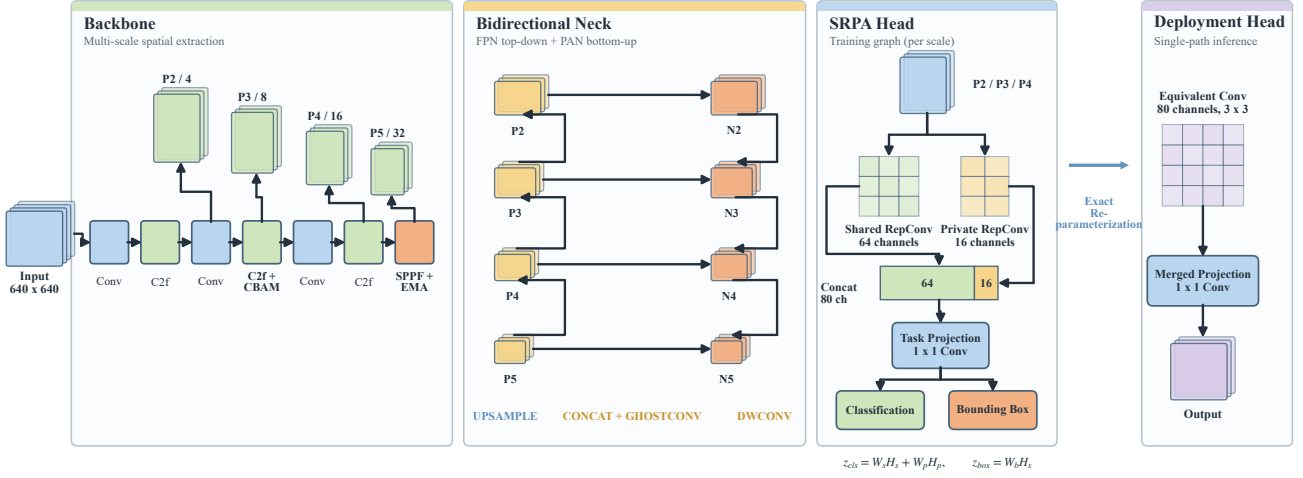


Fig. 1 Overall architecture of SRPA-YOLO. The multiscale backbone and bidirectional neck provide P2–P4 features. During training, shared and classification-private representations are optimized separately; exact fusion converts them into a single static deployment path

where K_l denotes the spatial convolution, BN denotes batch normalization, and $\mathcal{A}_l$ contains the SiLU activation and local C2f aggregation. The resulting P2, P3, P4, and P5 features have strides $s_l \in \{4, 8, 16, 32\}$. For a target whose transformed maximum side is d pixels, its approximate support on level l is

$$n_l = \frac{d}{s_l}. \quad (2)$$

Retaining P2 therefore gives a $d = 8$ pixel UAV approximately two feature cells instead of one cell at stride 8. This higher spatial sampling density reduces the probability that an extremely small target is eliminated before prediction.

3.2.2 Lightweight Bidirectional Feature Fusion

The neck combines top-down semantic propagation and bottom-up spatial aggregation. For adjacent pyramid levels, the top-down feature T_l and bottom-up feature N_l are

$$T_l = \psi_l([X_l, \text{Up}(T_{l+1})]), \quad N_l = \eta_l([T_l, \text{Down}(N_{l-1})]), \quad (3)$$

where $[\cdot, \cdot]$ denotes channel concatenation, Up is nearest-neighbor interpolation, Down is depthwise strided convolution, and ψ_l and η_l are compact C2f/Ghost feature mixers. The top-down path transfers semantic context to high-resolution maps, while the bottom-up path restores edge and displacement cues to deeper maps. This bidirectional exchange is particularly useful when a UAV has weak appearance but remains distinguishable through localized spatial contrast.

3.2.3 Context-Aware Feature Refinement

The high-resolution stage applies channel–spatial refinement, whereas the deepest stage aggregates a wider context. For a feature F , the refined response can be expressed as

$$\begin{aligned} \tilde{F} &= F \odot M_c(F) \odot M_s(F), \\ M_c(F) &= \sigma(g_c(\text{AvgPool} F, \text{MaxPool} F)), \\ M_s(F) &= \sigma(g_s(\text{Avg}_c F, \text{Max}_c F)), \end{aligned} \quad (4)$$

where $\odot$ is elementwise multiplication, σ is the sigmoid function, and g_c and g_s are lightweight channel and spatial transformations. At P2/P3, the spatial term suppresses cloud edges, foliage, and building textures that resemble tiny UAVs. At the deepest level, pooled contextual responses enlarge the effective receptive field and help distinguish an isolated aerial target from repetitive local clutter. These modules form the retained feature extractor; the contribution of this work concerns the task allocation inside the prediction head.

3.2.4 P2–P4 Prediction Interface

The bidirectional neck outputs N_l at $l \in \{2, 3, 4\}$. A scale-specific projection aligns each feature to the head input dimension,

$$F_l = P_l * N_l, \quad F_l \in \mathbb{R}^{C_l \times H_l \times W_l}, \quad (H_l, W_l) \in \{(160, 160), (80, 80), (40, 40)\}. \quad (5)$$

The three levels provide dense coverage for tiny and small UAVs while avoiding a runtime P5 prediction branch. The same SRPA allocation is applied independently at every level, allowing task separation without dynamic routing or scale-dependent control flow.

3.3 Shared Representation and Private-Rank Allocation

Conventional shared heads expose classification and localization to the same feature dictionary. SRPA instead retains a shared representation for both tasks and assigns a low-rank classification-private representation, as illustrated in Figure 2.

For input feature F_l , the shared and classification-private RepConv stems produce

$$H_l^s = \phi_s^{(l)}(F_l) \in \mathbb{R}^{64 \times H_l \times W_l}, \quad H_l^p = \phi_p^{(l)}(F_l) \in \mathbb{R}^{16 \times H_l \times W_l}. \quad (6)$$

Let z_l^{box} contain $4K$ DFL logits with $K = 16$ bins per coordinate and let z_l^{cls} contain the UAV class logit. SRPA computes

$$z_l^{box} = W_l^b H_l^s + b_l^b, \quad (7)$$

$$z_l^{cls} = W_l^s H_l^s + b_l^s + W_l^p H_l^p. \quad (8)$$

Here, the 64 and 16 channels are implementation-level representation dimensions, denoted by r_s and r_p below. They specify the available shared and private channel budgets; they do not assume that a learned convolution attains full algebraic rank. From Equations (7) and (8),

$$\frac{\partial z_l^{box}}{\partial H_l^p} = 0, \quad (9)$$

which establishes direct computational independence of the box output from the classification-private representation. Upstream task interaction remains mediated by the shared representation learned during Stage I. Flattening spatial dimensions gives $H_l^s \in \mathbb{R}^{r_s \times N_l}$ and $H_l^p \in \mathbb{R}^{r_p \times N_l}$, where $N_l = H_l W_l$. The class-logit row space satisfies

$$z_l^{cls} - b_l^{cls} \mathbf{1}^\top \in \text{row}(H_l^s) + \text{row}(H_l^p), \quad (10)$$

$$\dim(\text{row}(H_l^s) + \text{row}(H_l^p)) \leq r_s + r_p.$$

Equation (10) gives an upper bound on the combined row-space dimension; the attained dimension depends on the learned branch ranks and their subspace overlap. It shows that classification may access up to $r_s + r_p = 80$ representation dimensions, whereas localization is computed only from the $r_s = 64$ shared dimensions. The private class projection is initialized as $W_l^p = 0$, which, provided that the shared detector is unchanged at insertion, yields the function-preserving condition

$$f_{\text{SRPA}}(x; W^p = 0) = f_s(x), \quad \forall x. \quad (11)$$

The expanded detector therefore begins refinement from the trained shared detector with an identical prediction function. During private refinement, the backbone, neck, shared stems, box projections, and shared class projections remain fixed;

only ϕ_p and W^p are optimized. This constraint leaves the established localization mapping unchanged in Stage II while providing additional class-specific capacity for low-contrast UAVs.

3.4 Exact Structural Re-Parameterization

Each RepConv stem contains linear convolution and normalization branches during training. A convolution kernel K followed by batch normalization parameters $(\gamma, \beta, \mu, \sigma^2)$ is converted to

$$\hat{K} = \frac{\gamma}{\sqrt{\sigma^2 + \epsilon}} K, \quad \hat{b} = \beta + \frac{\gamma}{\sqrt{\sigma^2 + \epsilon}} (b - \mu). \quad (12)$$

After padding branch kernels to 3×3 , equivalent branch kernels and biases are summed. Denote the resulting shared and private stems by (K_s, b_s) and (K_p, b_p) , where $K_s \in \mathbb{R}^{64 \times C \times 3 \times 3}$ and $K_p \in \mathbb{R}^{16 \times C \times 3 \times 3}$. Their exact channel-wise fusion is

$$K_d = \begin{bmatrix} K_s \\ K_p \end{bmatrix}, \quad b_d = \begin{bmatrix} b_s \\ b_p \end{bmatrix}. \quad (13)$$

For $K = 16$ DFL bins and $n_c = 1$ class, the corresponding 1×1 projections have $W_b \in \mathbb{R}^{4K \times 64 \times 1 \times 1}$, $W_s \in \mathbb{R}^{n_c \times 64 \times 1 \times 1}$, and $W_p \in \mathbb{R}^{n_c \times 16 \times 1 \times 1}$. The deployed projections become

$$W_{box}^d = [W_b \ 0], \quad W_{cls}^d = [W_s \ W_p]. \quad (14)$$

For $H = [H_s; H_p]$, zero-padding the private columns of W_{box}^d gives $W_{box}^d H = W_b H_s$, while concatenating the class projections gives $W_{cls}^d H = W_s H_s + W_p H_p$. The training-time decomposition is therefore converted into one 80 -channel 3×3 convolution and one merged 1×1 projection per scale without changing the computed output apart from floating-point roundoff. Figure 3 summarizes this process.

Figure 4 details the single-scale head and its task-specific projections.

3.5 Objective Function and Two-Stage Optimization

The supervised objective is

$$\mathcal{L}_{sup} = \lambda_{box} \mathcal{L}_{CIoU} + \lambda_{cls} \mathcal{L}_{BCE} + \lambda_{dfl} \mathcal{L}_{DFL}, \quad (\lambda_{box}, \lambda_{cls}, \lambda_{dfl}) = (7.5, 0.5, 1.5), \quad (15)$$

where CIoU combines overlap, center-distance, and aspect-ratio penalties [22, 36]; BCE supervises class confidence; and DFL models each box coordinate as a discrete distribution.

A frozen RLRD-YOLO teacher regularizes the shared detector. With temperature $T = 2$, class-response distillation is

$$\mathcal{L}_{clsKD} = T^2 \text{KL}(\sigma(z_t^{cls}/T) \parallel \sigma(z_s^{cls}/T)). \quad (16)$$

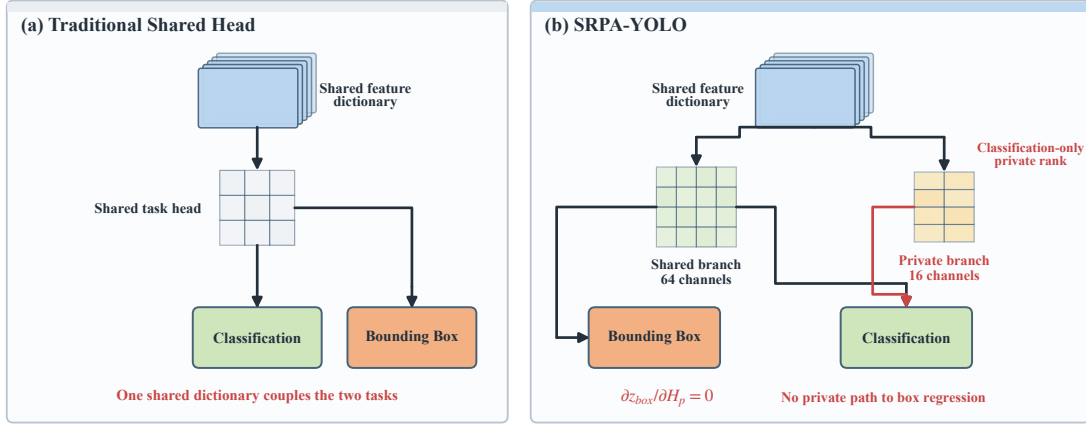


Fig. 2 Task-feature allocation in a conventional shared head and the proposed SRPA head. The private representation contributes only to classification, and its Jacobian to the box output is identically zero

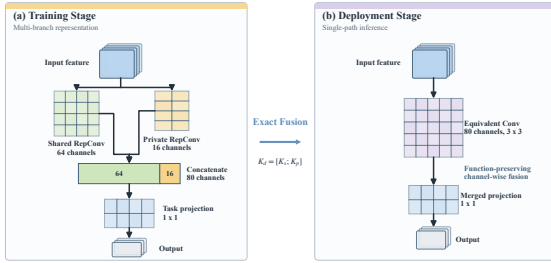


Fig. 3 Structural re-parameterization of the SRPA head. Parallel training representations provide task-specific capacity, while exact kernel and projection fusion produces a single-path deployment graph

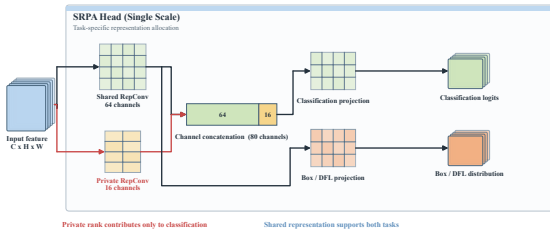


Fig. 4 Internal structure of the SRPA head. Shared features feed both output tasks, while the 16-channel private representation supplies an additive classification residual before deployment fusion

For positive locations Ω_+ , dense localization distillation and confidence-focused localization distillation are

$$\mathcal{L}_{dfIKD} = \frac{1}{N_+} \sum_{j \in \Omega_+} \text{KL}(P_t^{(j)} \| P_s^{(j)}), \quad (17)$$

$$\mathcal{L}_{focus} = \frac{\sum_{j \in \Omega_+} q_j^2 \text{KL}(P_t^{(j)} \| P_s^{(j)})}{\sum_{j \in \Omega_+} q_j^2 + \epsilon}, \quad (18)$$

where q_j is teacher confidence. The Stage-I objective is

$$\mathcal{L} = \mathcal{L}_{sup} + 0.25\mathcal{L}_{clsKD} + 0.50\mathcal{L}_{dfIKD} + 0.125\mathcal{L}_{focus}. \quad (19)$$

Stage II inserts the zero-output classification-private representation and optimizes only ϕ_p and W^p from the function-preserving state in Equation (11). The teacher remains frozen throughout distillation. This schedule first transfers localization knowledge to the compact shared detector and then refines class confidence while the Stage-II box function is fixed.

3.6 Experimental Environment and Training Settings

Detailed training parameters are provided in Table 1.

Table 1 Experimental environment and training settings

Item	Setting
Framework	Ultralytics 8.3.239 / PyTorch 2.9.0
CUDA	12.8
GPU	NVIDIA GeForce RTX 5060, 8 GB
Input size	640 × 640
Batch size	32
Epochs	200
Optimizer	SGD
Initial learning rate	0.01
Momentum	0.937
Weight decay	5×10^{-4}
Workers	2
Data augmentation	HSV jitter, horizontal flip, translation, scale, and mosaic

3.7 Metrics and Deployment Measurement

Precision and Recall are computed as

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}. \quad (20)$$

AP@0.50 and AP@0.75 integrate the precision–recall curve at IoU thresholds 0.50 and 0.75, respectively. The primary

metric averages AP over ten thresholds,

$$\text{mAP}_{0.50:0.95} = \frac{1}{10} \sum_{k=0}^9 \text{AP}_{0.50+0.05k}. \quad (21)$$

Evaluation uses batch size 32, confidence floor 0.001, NMS IoU 0.7, and at most 300 detections per image. Matched Stage-II experiments use seeds 40, 41, and 42 with the same Stage-I shared basis and independently initialized classification-private parameters to evaluate the stability of private refinement across different initializations.

For shared/private-width analysis, let m_c and a_c be validation mAP@0.50:0.95 and AP@0.75 for configuration c . Configurations within 0.003 of the highest validation mAP@0.50:0.95 form

$$\mathcal{C}_\varepsilon = \left\{ c : m_c \geq \max_j m_j - 0.003 \right\}, \quad (22)$$

and the member with the highest AP@0.75 is selected, with AP@0.50 resolving an exact tie. This rule preserves the primary metric while favoring more accurate localization.

Latency is measured at batch 1 and 640 pixels using FP16, fused convolution, forward inference plus NMS, 80 warm-up iterations, and 300 timed iterations. Each candidate and YOLOv8n are loaded in the same process and measured in alternating AB/BA order on the same GPU. Absolute latency and FPS are reported with the corresponding paired YOLOv8n control.

The real-time design is evaluated at the level of the complete detector path rather than by FLOPs alone. The measured frame time includes model forwarding, prediction decoding, and NMS; CUDA synchronization is applied around each timed iteration. Structural fusion is completed before timing, and no training-only classification-private path, dynamic gate, or auxiliary prediction remains in the deployed graph. The resulting latency is therefore the cost paid by the operational detector. Parameter count is reported together with latency because model storage and memory traffic constrain embedded execution even when arithmetic throughput is sufficient.

4 Experimental Results

4.1 Main Accuracy Comparison

Table 2 compares SRPA-YOLO with general-purpose compact detectors and UAV-oriented detectors on the deduplicated DUT Anti-UAV test set. SRPA-YOLO achieves 95.850% Precision, 92.201% Recall, 95.185% AP@0.50, 67.071% mAP@0.50:0.95, and 77.329% AP@0.75. YOLOv8n is the direct structural baseline because SRPA-YOLO retains its one-stage training and deployment framework. Relative to YOLOv8n, SRPA-YOLO improves the five metrics by 3.088, 8.242, 5.057, 6.895,

and 9.750 percentage points, respectively, while reducing deployed parameters by 0.842 M. Under the common evaluation protocol, the concurrent Precision and Recall gains make an explanation based solely on a shifted confidence operating point less plausible.

Figure 5 summarizes the accuracy and compactness comparison across the evaluated detectors.

YOLOv8n-P2 controls for the effect of adding a high-resolution prediction level. Although its Precision is 0.264 percentage points higher, SRPA-YOLO improves Recall, AP@0.50, mAP@0.50:0.95, and AP@0.75 by 1.826, 1.208, 1.664, and 1.039 percentage points, respectively. This comparison shows that task-feature allocation inside the detection head contributes additional gains beyond high-resolution scale extension.

Figure 6 relates deployed parameters to the primary accuracy metric. SRPA-YOLO occupies the upper-left region, combining the highest measured mAP@0.50:0.95 with the smallest parameter count among the compared detectors. This operating point is consistent with the design objective of assigning a compact shared dictionary and limiting additional capacity to the classification task.

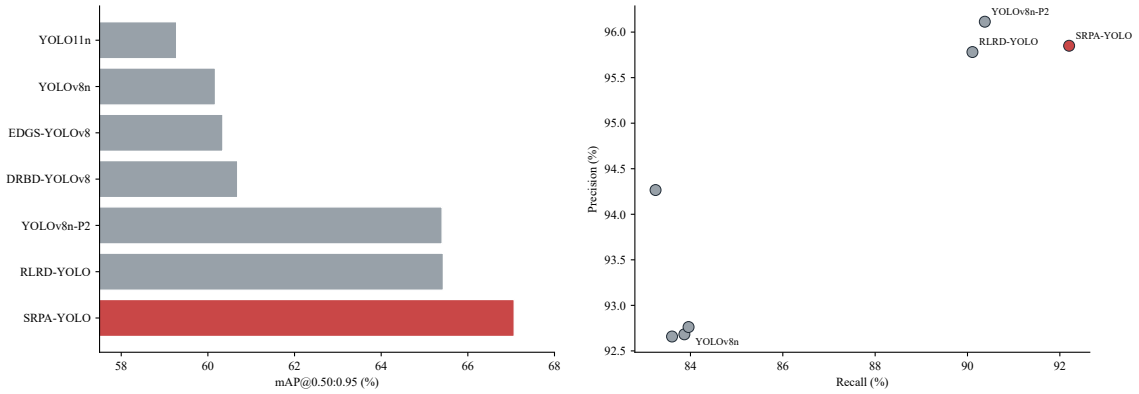
4.2 Shared/Private-Rank and Mechanism Ablations

Table 3 isolates the effect of shared and private ranks on the validation set. Increasing the shared rank from 48 to 64 or 80 does not lead to a monotonic increase in mAP@0.50:0.95, indicating that uniform width expansion is not sufficient to determine the optimum. The 48+16 allocation reaches the highest validation mAP@0.50:0.95 (60.874%), whereas 64+16 produces the highest AP@0.75 (69.986%), AP@0.50 (91.850%), and Precision (96.615%). According to Equation (22), both allocations lie within 0.3 percentage points of the highest mAP@0.50:0.95, and 64+16 is selected because it provides the strongest high-IoU localization.

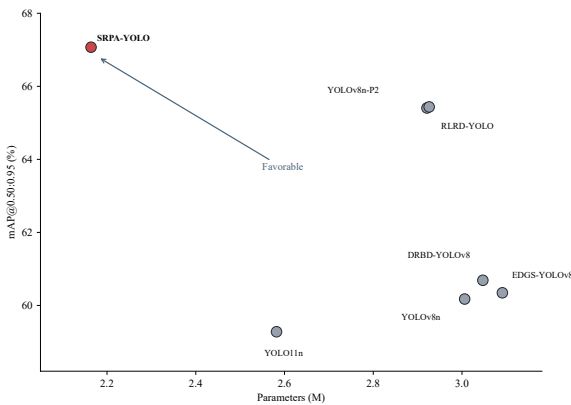
Table 4 traces cumulative changes from a nonlinear head to the final SRPA configuration. Adding dense DFL distillation is associated with recovery from the accuracy loss of head compression, and the subsequent confidence-focused term adds 0.384 mAP@0.50:0.95 points. Under the cumulative protocol, transferred initialization coincides with increases of 2.660 Recall points and 0.678 mAP@0.50:0.95 points; increasing the shared representation to 64 channels is followed by gains of 0.531 mAP@0.50:0.95 points and 1.674 AP@0.75 points. The final 16-channel classification-private representation reaches 67.071% mAP@0.50:0.95 and 77.329% AP@0.75. Each row retains the preceding components and therefore reports the performance of the corresponding cumulative configuration. The shared-80 control reaches a similar mAP@0.50:0.95 of 67.121%, whereas the 64+16 allocation additionally enforces the zero box Jacobian

Table 2 Comparison on 2199 deduplicated DUT Anti-UAV test images. Detection metrics are reported in percent; best values are bold

Method	Params (M)	Precision (%)	Recall (%)	AP@0.50 (%)	mAP@0.50:0.95 (%)	AP@0.75 (%)
YOLO11n	2.582	92.682	83.868	90.100	59.280	66.383
YOLOv8n	3.006	92.762	83.959	90.128	60.176	67.579
EDGS-YOLOv8	3.091	94.266	83.244	90.428	60.347	66.415
DRBD-YOLOv8	3.047	92.658	83.601	90.203	60.689	67.459
YOLOv8n-P2	2.921	96.114	90.375	93.977	65.407	76.290
RLRD-YOLO	2.926	95.781	90.107	94.467	65.436	74.757
SRPA-YOLO	2.164	95.850	92.201	95.185	67.071	77.329
Δ SRPA vs. YOLOv8n	-0.842	+3.088	+8.242	+5.057	+6.895	+9.750

**Fig. 5** Accuracy and compactness comparison on the deduplicated DUT Anti-UAV test set. All detection metrics are reported in percent.**Table 3** Validation-selected shared/private rank ablation. Detection metrics are percentages; $r_s + r_p$ denotes shared rank plus classification-private rank

r_s	r_p	Params (M)	Precision (%)	Recall (%)	AP@0.50 (%)	mAP@0.50:0.95 / AP@0.75 (%)
48	0	2.093	96.089	87.182	91.667	60.508 / 69.337
64	0	2.128	96.378	86.570	91.703	60.454 / 69.754
80	0	2.164	96.073	87.743	91.569	60.513 / 69.262
48	16	2.128	96.135	87.310	91.798	60.874 / 69.618
64	16	2.164	96.615	86.570	91.850	60.646 / 69.986

**Fig. 6** Model-complexity trade-off. The horizontal axis is deployed parameter count and the vertical axis is mAP@0.50:0.95.

in Equation (9) and is selected by the predefined validation localization criterion. Together with the rank ablation, the results support task-constrained capacity allocation instead of unrestricted head widening.

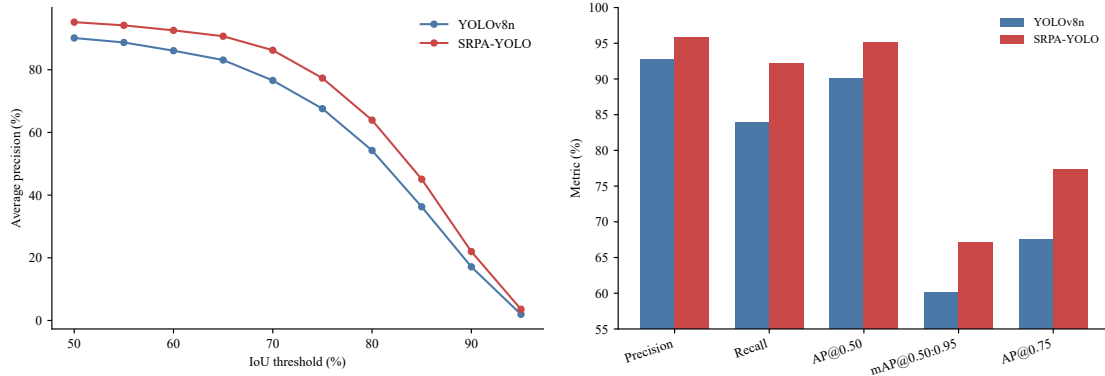
4.3 Localization and Scale Characteristics

Figure 7 compares SRPA-YOLO directly with YOLOv8n over IoU thresholds from 0.50 to 0.95. SRPA-YOLO has higher AP at every threshold. The margin is 5.057 percentage points at IoU 0.50, increases to 9.750 points at IoU 0.75, and remains 8.803 points at IoU 0.85. The persistence of the margin under stricter overlap criteria indicates that the measured improvement includes a localization component and is not confined to class confidence at IoU 0.50.

The largest aggregate improvement over YOLOv8n is observed in Recall (+8.242 points), accompanied by gains in both AP@0.50 and AP@0.75. In long-range UAV monitoring, improved Recall directly reduces missed targets, while the concurrent Precision increase limits additional false alarms. The P2 prediction level provides spatial support for tiny UAVs, while the SRPA head keeps classification-private refinement outside the box-regression input. This structural

Table 4 Mechanism ablation on the deduplicated DUT Anti-UAV test set. Detection metrics are percentages

Configuration	Params (M)	Precision (%)	Recall (%)	AP@0.50 (%)	mAP@0.50:0.95 (%)	AP@0.75 (%)
Full nonlinear head	2.603	95.922	90.954	94.736	66.046	75.503
Shared-48 + dense DFL KD	2.093	96.152	90.954	94.836	65.363	74.929
+ focused localization KD	2.093	97.084	90.285	95.018	65.747	75.573
+ transferred initialization	2.093	96.484	92.945	95.493	66.425	75.472
+ shared rank 64	2.128	96.122	92.157	95.366	66.956	77.146
+ private rank 16	2.164	95.850	92.201	95.185	67.071	77.329

**Fig. 7** Localization comparison between SRPA-YOLO and YOLOv8n. Left: AP at individual IoU thresholds; right: aggregate detection metrics.

interpretation is consistent with the observed gains at both low and high IoU thresholds.

4.4 Qualitative Detection Analysis

Figure 8 compares YOLOv8n and SRPA-YOLO on the same tiny-UAV instances at a confidence threshold of 0.25. A detection is regarded as correct when its IoU with the ground-truth box is at least 0.50. In all three selected examples, YOLOv8n produces no matched prediction, whereas SRPA-YOLO recovers the target. These representative cases visually corroborate the quantitative improvements reported on the test set. The enlarged regions show the spatial alignment of the recovered predictions for targets represented by only a few pixels.

Figure 9 further presents successful SRPA-YOLO detections under sky, building, vegetation, and low-contrast backgrounds. The selected targets have projected longest-edge lengths of 5.7–6.3 pixels after 640-pixel letterboxing. Despite the limited target support and substantial surrounding background, the final validation-selected checkpoint produces localized responses in each selected scene. These cases illustrate the model’s small-target detection behavior and are consistent with the preceding test-set metrics.

4.5 Matched-Seed Stability

Table 5 summarizes the results for Stage-II seeds 40, 41, and 42. All experiments use the same Stage-I shared basis and

independently initialize the classification-private parameters. SRPA-YOLO yields standard deviations of 0.112, 0.022, and 0.082 percentage points for Recall, AP@0.50, and mAP@0.50:0.95, respectively, indicating low variation across these private-parameter initializations. Relative to RLRD-YOLO, the mean paired differences in Recall, mAP@0.50:0.95, and AP@0.75 are +2.709, +1.881, and +2.150 percentage points, respectively. Under the matched protocol, the observed improvement is consistent across the three Stage-II seeds.

Table 5 Matched Stage-II seeds on the deduplicated DUT Anti-UAV test set (percent, mean $\pm$ sample standard deviation, $n = 3$). All runs use the same Stage-I shared basis and independently initialize only the private parameters

Metric (%)	RLRD-YOLO	SRPA-YOLO	Paired diff.
Precision	95.964 $\pm$ 0.766	96.167 $\pm$ 0.031	+0.204 $\pm$ 0.736
Recall	89.463 $\pm$ 0.785	92.172 $\pm$ 0.112	+2.709 $\pm$ 0.773
AP@0.50	94.661 $\pm$ 0.106	95.304 $\pm$ 0.022	+0.644 $\pm$ 0.109
mAP@0.50:0.95	65.199 $\pm$ 0.639	67.080 $\pm$ 0.082	+1.881 $\pm$ 0.721
AP@0.75	75.196 $\pm$ 0.564	77.345 $\pm$ 0.233	+2.150 $\pm$ 0.332

4.6 Latency and Deployment Complexity

The training graph contains approximately 2.18 M parameters and requires 8.5 GFLOPs at 640 pixels. Structural fusion removes training-only state and produces a 2.164 M-parameter checkpoint. Table 6 reports same-process paired

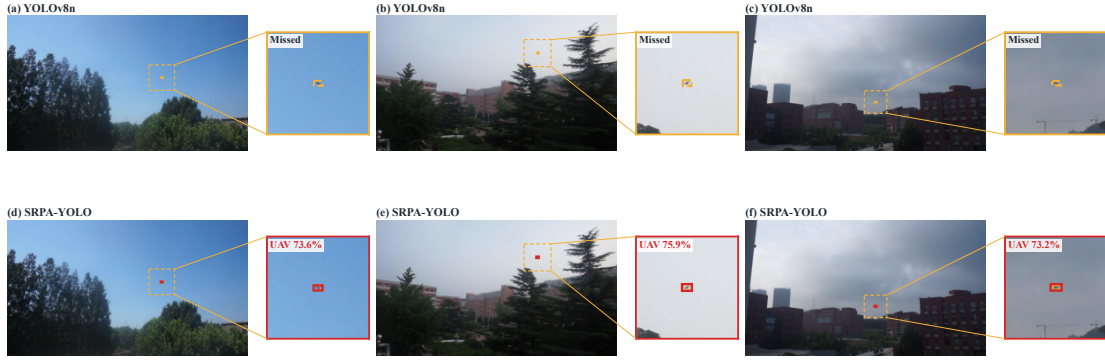


Fig. 8 Matched-image detection comparison on the deduplicated DUT Anti-UAV test set. Yellow dashed boxes denote ground truth, red solid boxes denote SRPA-YOLO predictions, and “Missed” indicates that YOLOv8n has no prediction satisfying confidence ≥ 0.25 and IoU ≥ 0.50 .

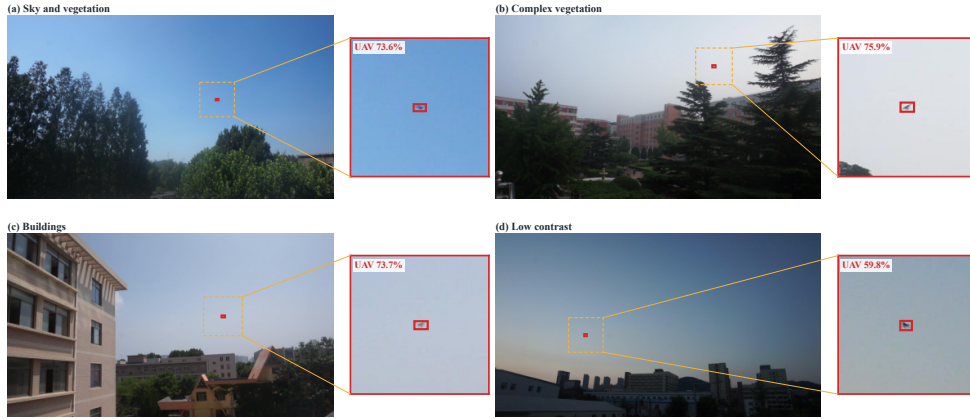


Fig. 9 Successful SRPA-YOLO detections of tiny UAVs. Each right-hand view enlarges the target region from the corresponding full image; red boxes and labels are predictions from the final validation-selected checkpoint.

Table 6 Same-process paired latency. Each row reports a candidate and its paired YOLOv8n control; batch 1, 640 pixels, FP16, fused forward+NMS, 80 warm-ups, and 300 iterations

Method	Round	ms	FPS	Control ms
YOLO11n	1	21.6357	46.22	16.8632
	2	19.1505	52.22	15.0139
YOLOv8n-P2	1	19.1953	52.10	15.9227
	2	16.9629	58.95	14.1962
EDGS*	1	19.5146	51.24	14.6088
	2	22.0043	45.45	16.3604
DRBD*	1	29.0158	34.46	14.5733
	2	31.7115	31.53	16.8584
RLRD-YOLO	1	22.5644	44.32	16.0771
	2	22.7773	43.90	16.2647
SRPA-YOLO	1	16.2469	61.55	16.4922
	2	15.0936	66.25	15.5909

measurements, in which each detector is evaluated together with a paired YOLOv8n control; Table 7 gives the separate low-load final-acceptance measurements.

In the separate low-load acceptance measurements, SRPA-YOLO requires 9.7435 and 9.8574 ms (102.63 and 101.45 FPS), whereas the paired YOLOv8n control requires 9.8859 and 9.9893 ms (101.15 and 100.11 FPS). The two-round means are 9.8005 ms and 9.9376 ms, respectively. Under the reported protocol, SRPA-YOLO achieves com-

Table 7 Separate low-load final-acceptance measurements for SRPA-YOLO and its paired YOLOv8n control under the same batch-1, 640-pixel, FP16, fused forward+NMS protocol

Round	SRPA ms	SRPA FPS	YOLOv8n ms	YOLOv8n FPS
1	9.7435	102.63	9.8859	101.15
2	9.8574	101.45	9.9893	100.11

parable frame time while reducing the deployed parameter count by 28.0%.

Figure 10 combines the reported two-round mean latency with the primary accuracy metric. Among the measured models and under this hardware protocol, SRPA-YOLO occupies the favorable accuracy-speed boundary: models with comparable accuracy have higher measured latency, whereas faster models do not reach the same mAP@0.50:0.95.

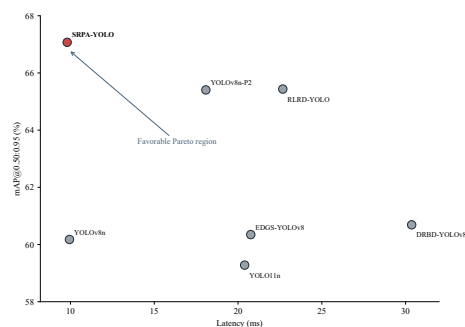


Fig. 10 Accuracy–speed trade-off based on the reported two-round mean latency and mAP@0.50:0.95.

5 Conclusions

This paper presented SRPA-YOLO, a compact detector that formulates prediction-head capacity as task-specific representation allocation. A 64-channel shared representation supports classification and DFL localization, whereas a zero-output-initialized 16-channel classification-private representation contributes only an additive class residual. This construction increases the feature capacity available to classification while making the box output directly independent of the private representation. Exact structural re-parameterization converts the training-time decomposition into one 80-channel convolution and one merged output projection per scale, yielding a static single-path deployment graph.

On the deduplicated DUT Anti-UAV test set, the selected checkpoint achieves 95.850% Precision, 92.201% Recall, 95.185% AP@0.50, 67.071% mAP@0.50:0.95, and 77.329% AP@0.75 with 2.164 M deployed parameters. Compared with YOLOv8n, the corresponding improvements are 3.088, 8.242, 5.057, 6.895, and 9.750 percentage points, together with 28.0% fewer parameters. The additional gains in Recall, AP@0.50, mAP@0.50:0.95, and AP@0.75 over YOLOv8n-P2 show that head-level task allocation complements high-resolution prediction. The gains across the evaluated IoU thresholds and the low variation in the matched-seed experiments are consistent with stable localization and classification refinement under the reported protocol.

Two paired deployment measurements yield 9.7435 and 9.8574 ms for SRPA-YOLO and 9.8859 and 9.9893 ms for the paired YOLOv8n control under the specified RTX 5060, FP16, batch-1, forward-plus-NMS protocol. Together with the 28.0% parameter reduction, these measurements indicate a favorable accuracy–compactness operating point on the evaluated GPU software stack. Future work should examine multi-class aerial scenes, embedded ONNX/TensorRT deployment, and scale-conditioned private capacity under additional target-size and imaging conditions.

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Author contributions. Xianpu Wang: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, and writing—original draft. Haijie Zhi: validation, formal analysis, and writing—review and editing. Qing Fei: conceptualization, supervision, resources, project administration, and writing—review and editing. All authors read and approved the final manuscript.

Data and code availability. The DUT Anti-UAV dataset is available from its public project repository subject to the provider’s terms. The reproducibility package contains the deduplication manifest, source code, model configurations, result manifest, paired-latency records, and editable plotting scripts. These materials and the trained weights are available from the corresponding author and will be deposited in a versioned public repository. One cross-split duplicate is excluded by a fixed manifest and documented by its SHA-256 hash; no source image is deleted.

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References

1. Bo, C., Wei, Y., Wang, X., et al.: Vision-based anti-uav detection based on yolov7-gs in complex backgrounds. *Drones* **8**(7), 331 (2024). DOI 10.3390/drones8070331
2. Bochkovskiy, A., Wang, C.Y., Liao, H.Y.M.: Yolov4: Optimal speed and accuracy of object detection. arXiv preprint arXiv:2004.10934 (2020). DOI 10.48550/arXiv.2004.10934
3. Chen, J., Kao, S.H., He, H., Zhuo, W., Wen, S., Lee, C.H., Chan, S.H.G.: Run, don't walk: Chasing higher flops for faster neural networks. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 12021–12031 (2023). DOI 10.1109/CVPR52729.2023.01157
4. Ding, X., Zhang, X., Ma, N., Han, J., Ding, G., Sun, J.: Repvgg: Making vgg-style convnets great again. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 13733–13742 (2021). DOI 10.1109/CVPR46437.2021.01352
5. Feng, C., Zhong, Y., Gao, Y., Scott, M.R., Huang, W.: Toood: Task-aligned one-stage object detection. In: *Proc. IEEE/CVF International Conference on Computer Vision*, pp. 3490–3499 (2021). DOI 10.1109/ICCV48922.2021.00348
6. Han, K., Wang, Y., Tian, Q., Guo, J., Xu, C., Xu, C.: Ghostnet: More features from cheap operations. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 1580–1589 (2020). DOI 10.1109/CVPR42600.2020.00165
7. Hinton, G., Vinyals, O., Dean, J.: Distilling the knowledge in a neural network. arXiv preprint arXiv:1503.02531 (2015). DOI 10.48550/arXiv.1503.02531

8. Howard, A.G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M., Adam, H.: Mobilenets: Efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861* (2017). DOI 10.48550/arXiv.1704.04861
9. Huang, B., Li, J., Chen, J., Wang, G., et al.: Anti-uav410: A thermal infrared benchmark and customized scheme for tracking drones in the wild. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **46**(5), 2852–2865 (2024). DOI 10.1109/TPAMI.2023.3335338
10. Huang, M., Mi, W., Wang, Y.: Edgs-yolov8: An improved yolov8 lightweight uav detection model. *Drones* **8**(7), 337 (2024). DOI 10.3390/drones8070337
11. Jiang, N., Wang, K., Peng, X., Yu, X., Wang, Q., Xing, J., Li, G., Guo, Q., Hu, Z., Han, Z., et al.: Anti-uav: A large-scale benchmark for vision-based uav tracking. *IEEE Transactions on Multimedia* **25**, 486–500 (2023). DOI 10.1109/TMM.2021.3128047
12. Jiang, P., Yang, X., Wan, Y., et al.: Drbd-yolov8: A lightweight and efficient anti-uav detection model. *Sensors* **24**(22), 7148 (2024). DOI 10.3390/s24227148
13. Li, H., Li, Y., Xiao, L., et al.: Rlrd-yolo: An improved yolov8 algorithm for small object detection from an unmanned aerial vehicle perspective. *Drones* **9**(4), 293 (2025). DOI 10.3390/drones9040293
14. Li, X., Wang, W., Wu, L., Chen, S., Hu, X., Li, J., Tang, J., Yang, J.: Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection. In: *Advances in Neural Information Processing Systems*, vol. 33, pp. 21002–21012 (2020)
15. Lin, T.Y., Dollár, P., Girshick, R., He, K., Hariharan, B., Belongie, S.: Feature pyramid networks for object detection. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 2117–2125 (2017). DOI 10.1109/CVPR.2017.106
16. Lin, T.Y., Goyal, P., Girshick, R., He, K., Dollár, P.: Focal loss for dense object detection. In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 2999–3007 (2017). DOI 10.1109/ICCV.2017.324
17. Lin, T.Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.L.: Microsoft coco: Common objects in context. In: *European Conference on Computer Vision*, pp. 740–755 (2014). DOI 10.1007/978-3-319-10602-1_48
18. Liu, S., Qi, L., Qin, H., Shi, J., Jia, J.: Path aggregation network for instance segmentation. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 8759–8768 (2018). DOI 10.1109/CVPR.2018.00913
19. Liu, W., Anguelov, D., Erhan, D., Szegedy, C., Reed, S., Fu, C.Y., Berg, A.C.: Ssd: Single shot multibox detector. In: *European Conference on Computer Vision*, pp. 21–37 (2016). DOI 10.1007/978-3-319-46448-0_2
20. Ouyang, D., He, S., Zhang, G., Luo, M., Guo, H., Zhan, J., Huang, Z.: Efficient multi-scale attention module with cross-spatial learning. In: *IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 1–5 (2023). DOI 10.1109/ICASSP49357.2023.10096516
21. Redmon, J., Divvala, S., Girshick, R., Farhadi, A.: You only look once: Unified, real-time object detection. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 779–788 (2016). DOI 10.1109/CVPR.2016.91
22. Rezatofighi, H., Tsoi, N., Gwak, J., Sadeghian, A., Reid, I., Savarese, S.: Generalized intersection over union: A metric and a loss for bounding box regression. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* pp. 658–666 (2019). DOI 10.1109/CVPR.2019.00075
23. Tan, M., Pang, R., Le, Q.V.: Efficientdet: Scalable and efficient object detection. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10781–10790 (2020). DOI 10.1109/CVPR42600.2020.01079
24. Tian, Z., Shen, C., Chen, H., He, T.: Fcos: Fully convolutional one-stage object detection. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 9627–9636 (2019). DOI 10.1109/ICCV.2019.00972
25. Tong, K., Wu, Y., Zhou, F.: Recent advances in small object detection based on deep learning: A review. *Image and Vision Computing* **97**, 103910 (2020). DOI 10.1016/j.imavis.2020.103910
26. Ultralytics: Ultralytics yolo (2023). URL <https://github.com/ultralytics/ultralytics>. Version 8.3.239; accessed 18 July 2026
27. Wang, C.Y., Bochkovskiy, A., Liao, H.Y.M.: Yolo7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 7464–7475 (2023). DOI 10.1109/CVPR52729.2023.00721
28. Woo, S., Park, J., Lee, J.Y., Kweon, I.S.: Cbam: Convolutional block attention module. In: *European Conference on Computer Vision*, pp. 3–19 (2018). DOI 10.1007/978-3-030-01234-2_1
29. Wu, Y., Chen, Y., Yuan, L., Liu, Z., Wang, L., Li, H., Fu, Y.: Rethinking classification and localization for object detection. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10186–10195 (2020). DOI 10.1109/CVPR42600.2020.01020
30. Yang, H., Liang, B., Feng, S., Jiang, J., Fang, A., Li, C.: Lightweight uav detection method based on iasl-yolo. *Drones* **9**(5), 325 (2025). DOI 10.3390/drones9050325
31. Yang, Z., Li, Z., Jiang, X., Gong, Y., Yuan, Z., Zhao, D., Yuan, C.: Focal and global knowledge distillation for detectors. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 4643–4652 (2022). DOI 10.1109/CVPR52688.2022.00461
32. Yasmeen, A., Daescu, O.: Recent research progress on ground-to-air vision-based anti-uav detection and tracking methodologies: A review. *Drones* **9**(1), 58 (2025). DOI 10.3390/drones9010058
33. Zagoruyko, S., Komodakis, N.: Paying more attention to attention: Improving the performance of convolutional neural networks via attention transfer. In: *International Conference on Learning Representations* (2017)
34. Zhang, Q., Wang, X., Shi, H., Wang, K., Tian, Y., Xu, Z., Zhang, Y., Jia, G.: Bra-yolov10: A lightweight small object detection model for uav images. *Drones* **9**(3), 159 (2025). DOI 10.3390/drones9030159
35. Zhao, J., Zhang, J., Li, D., Wang, D., et al.: Vision-based anti-uav detection and tracking. *IEEE Transactions on Intelligent Transportation Systems* **23**(12), 25323–25334 (2022). DOI 10.1109/TITS.2022.3182897
36. Zheng, Z., Wang, P., Liu, W., Li, J., Ye, R., Ren, D.: Distance-iou loss: Faster and better learning for bounding box regression. *Proceedings of the AAAI Conference on Artificial Intelligence* **34**(7), 12993–13000 (2020). DOI 10.1609/aaai.v34i07.6999
37. Zheng, Z., Ye, R., Wang, P., Ren, D., Zuo, W., Hou, Q., Cheng, M.M.: Localization distillation for dense object detection. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 9407–9416 (2022). DOI 10.1109/CVPR52688.2022.00919
38. Zhu, P., Wen, L., Du, D., Bian, X., Fan, H., Hu, Q., Ling, H.: Vision meets drones: A challenge. *arXiv preprint arXiv:1804.07437* (2018). DOI 10.48550/arXiv.1804.07437